Rules of Engagement:

- You are *not* allowed to work with your peers.
- Show all work and derivations to receive credit.
- Write clearly and legibly for your own benefit.
- You are *not* allowed to use chatGPT or other chatbots for this exam; you are also not allow a caluclator, phone or any other electronic devices.
- You are only allowed to use the provided ”cheat sheet” and scrap paper (which you will turn in at the end of the exam).

Name : ______________________________

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Problem 1 (k-means: Mean-square distance expansion identity) [6pts].

Let $x_1, \ldots, x_L \in \mathbb{R}^n$ and define $\bar{x} = \frac{1}{L}\sum_i x_i$. Show that for any z,

$$J(z) = \sum_{i=1}^{L} \|x_i - z\|^2 = \sum_{i=1}^{L} \|x_i - \bar{x}\|^2 + L\|z - \bar{x}\|^2.$$

Use this identity to show that the minimum of the k-means objective is achieved when $z = \bar{x}$.

Note that we saw this identity in class/homework, and it is called the **"centering identity" or "bias–variance" style decomposition**, and explains algebraically why the mean minimizes total squared distance. **Solution.**

Expand:

$$\sum_i \|x_i - z\|^2 = \sum_i \|(x_i - \bar{x}) - (z - \bar{x})\|^2 = \sum_i (\|x_i - \bar{x}\|^2 - 2(x_i - \bar{x})^\top (z - \bar{x}) + \|z - \bar{x}\|^2).$$

The cross term sums to zero because $\sum_i (x_i - \bar{x}) = 0$. Hence

$$J(z) = \sum_i \|x_i - \bar{x}\|^2 + L\|z - \bar{x}\|^2.$$

This shows $J(z) \geq J(\bar{x})$, with equality only when $z = \bar{x}$. Therefore, the minimum of the k-means objective is achieved when $z = \bar{x}$.

Problem 2 (Minimizing weighted mean-square distance) [6pts].

Let $x_1, \ldots, x_L \in \mathbb{R}^n$ and weights $w_1, \ldots, w_L > 0$ with $\sum_{i=1}^{L} w_i = 1$. Consider

$$J(z) \;=\; \sum_{i=1}^{L} w_i \, \|x_i - z\|_2^2, \qquad z \in \mathbb{R}^n.$$

(a) Show that J is a quadratic in z and identify its Hessian. Conclude that J is strictly convex and hence has a unique minimizer.

(b) Find the unique minimizer z^* of $J(z)$ in closed form.

Solution.

(a) **Convexity and Hessian.** Expand each term:

$$\|x_i - z\|_2^2 \;=\; (x_i - z)^\top (x_i - z) \;=\; \|x_i\|_2^2 - 2x_i^\top z + \|z\|_2^2.$$

Therefore

$$J(z) = \sum_{i=1}^{L} w_i \big(\|x_i\|_2^2 - 2x_i^\top z + \|z\|_2^2\big) = \underbrace{\sum_{i=1}^{L} w_i \|x_i\|_2^2}_{\text{constant in } z} - 2\Big(\sum_{i=1}^{L} w_i x_i\Big)^\top z + \Big(\sum_{i=1}^{L} w_i\Big)\|z\|_2^2.$$

Since $\sum_i w_i = 1$, we have

$$J(z) \;=\; \text{const} - 2\Big(\sum_{i=1}^{L} w_i x_i\Big)^\top z + \|z\|_2^2.$$

Thus $J(z)$ is a quadratic function with gradient and Hessian

$$\nabla J(z) \;=\; -2\sum_{i=1}^{L} w_i x_i + 2z, \qquad \nabla^2 J(z) \;=\; 2I_n.$$

The Hessian $2I_n$ is positive definite, hence J is *strictly convex*, so it has a unique minimizer.

(b) **Minimizer and closed form.** Set the gradient to zero:

$$\nabla J(z) = 0 \quad\Longleftrightarrow\quad -2\sum_{i=1}^{L} w_i x_i + 2z = 0 \quad\Longleftrightarrow\quad z^* \;=\; \sum_{i=1}^{L} w_i x_i.$$

This is the unique minimizer by strict convexity.

Problem 3 (Projection Matrices) [6pts]. Let $a \in \mathbb{R}^n$ be a nonzero vector. We define the projection of any $x \in \mathbb{R}^n$ onto the line spanned by a as

$$P_a x := \frac{aa^\mathsf{T}}{a^\mathsf{T}a}\, x.$$

(a) Compute P explicitly for $a = \begin{bmatrix}2\\1\end{bmatrix}$.

(b) Show that P is symmetric (i.e., $P = P^\top$)and idempotent (i.e., $P^2 = P$).

(c) If $Q := I - P$, show that Q is also symmetric and idempotent, and that $Qx = x - Px$ gives the projection of x onto the orthogonal complement of a.

(d) For $a = \begin{bmatrix}2\\1\end{bmatrix}$ and $x = \begin{bmatrix}1\\3\end{bmatrix}$, compute Px and Qx and verify that $x = Px + Qx$ with $Px \perp Qx$.

Solution.

(a) $P = \frac{1}{5}\begin{bmatrix}4 & 2\\ 2 & 1\end{bmatrix}$.

(b) $P^\mathsf{T} = P$ because $(aa^\mathsf{T})^\mathsf{T} = aa^\mathsf{T}$. Also $P^2 = \frac{(aa^\mathsf{T})(aa^\mathsf{T})}{(a^\mathsf{T}a)^2} = \frac{a(a^\mathsf{T}a)a^\mathsf{T}}{(a^\mathsf{T}a)^2} = P$.

(c) $Q = I - P$ satisfies $Q^\mathsf{T} = Q$ and $Q^2 = Q$. Moreover $Qx = x - Px$ and $P^\mathsf{T}Q = 0$, so Px and Qx are orthogonal.

(d) With $a = (2, 1)$ and $x = (1, 3)$,

$$Px = \tfrac{1}{5}\begin{bmatrix}4 & 2\\ 2 & 1\end{bmatrix}\begin{bmatrix}1\\3\end{bmatrix} = \tfrac{1}{5}\begin{bmatrix}10\\5\end{bmatrix} = \begin{bmatrix}2\\1\end{bmatrix}, \qquad Qx = \begin{bmatrix}1\\3\end{bmatrix} - \begin{bmatrix}2\\1\end{bmatrix} = \begin{bmatrix}-1\\2\end{bmatrix}.$$

Their dot product is $(-1)(2) + 2(1) = 0$, confirming orthogonality.

Problem 4 (Projection onto a Line in Data Space) [6pts].

Given $x, y \in \mathbb{R}^m$, find $\alpha \in \mathbb{R}$ minimizing $\|y - \alpha x\|_2^2$. Show that the fitted value is the orthogonal projection of y onto $\mathrm{span}\{x\}$.

Solution. Expanding the cost we have that

$$J(\alpha) = \|y - \alpha x\|^2 = y^\top y - 2\alpha x^\top y + \alpha^2 x^\top x.$$

Taking the first derivative and setting it to zero we have that

$$J'(\alpha) = -2x^\top y + 2\alpha x^\top x = 0 \implies \alpha = \frac{x^\top y}{x^\top x}.$$

Thus $\hat{y} = \hat{\alpha} x = \frac{xx^\top}{x^\top x} y =: P_x y$, where P_x is symmetric idempotent and projects onto $\mathrm{span}\{x\}$.

Problem 5 (Nearest Unit-Vector (Projection onto the Unit Sphere)) [6pts]. Find $\hat{x}$ minimizing $\|y - \hat{x}\|_2^2$ subject to $\|x\|_2 = 1$.

Solution.

$$\|y - x\|^2 = \|y\|^2 - 2y^\top x + \|x\|^2 = \|y\|^2 - 2y^\top x + 1.$$

Since $\|x\| = 1$ and the only term in the above expression that depends on x is $y^\top x$, minimizing is equivalent to maximizing $y^\top x$ over $\|x\| = 1$. The maximum is achieved when x is in the direction of y, so the maximizer is $x = \frac{y}{\|y\|}$ (Cauchy–Schwarz). Hence

$$\hat{x} = \frac{y}{\|y\|}.$$

Geometrically, it is the orthogonal (radial) projection of y onto the unit sphere.